

Dual Temporal Masked Modeling for KPI Anomaly Detection via Similarity Aggregation

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Abstract—With the expanding scale of current industries, monitoring systems centered around Key Performance Indicators (KPIs) play an increasingly crucial role. KPI anomaly detection can monitor the potential risks according to KPI data and has garnered widespread attention due to its rapid responsiveness and adaptability to dynamic changes. Considering the absence of labels and the high cost of manual annotation of KPI data, the self-supervised approaches are proposed. Among them, mask modeling methods draw great attention and can learn the intrinsic distribution of data without relying on prior assumptions. However, conventional mask modeling often overlooks the examination of relationships between unsynchronized variables, treating them with equal importance, and inducing inaccurate detection results. To address this, this paper proposes a Dual Masked modeling Approach combined with Similarity Aggregation, named DMASA. Starting from a self-supervised approach based on mask modeling, DMASA incorporates spectral residual techniques to explore inter-variable dependencies and aggregates information from similar data to eliminate interference from irrelevant variables in anomaly detection. Extensive experiments on eight datasets and state-of-the-art results demonstrate the effectiveness of our approach. Our code is available at <https://github.com/colaudiolab/GT-DMASA>.

Index Terms—Anomaly detection, key performance indicators, self-supervised learning, monitoring.

I. INTRODUCTION

IN CONTEMPORARY industrial landscape, the continuous expansion of infrastructure, and the proliferation of sensors underscore the significance of monitoring systems. Employing monitoring systems is crucial for ensuring the continuous and reliable operation of diverse systems, including financial systems [1], [2], Internet-based online service systems [3], [4], and critical infrastructures such as network systems [5], [6] and water treatment facilities [7]. For example, Key Performance Indicators (KPIs), in the context of network security, encompass a wide array of data points, such as login failures, unusual data transfers, and unauthorized access attempts, which are vital for detecting anomalies that could signify security breaches or system irregularities. This

is essential for preventing potential risks, mitigating economic losses, and maintaining overall safety [8]. Common monitoring techniques encompass the utilization of KPIs and the analysis of logs and trace data. Considering the dynamic nature and time sensitivity of these systems, many opt for monitoring strategies centered around KPIs.

KPIs, often time series data, face anomaly detection challenges due to scarce labeled data and high labeling costs in large-scale systems [9]. Consequently, many KPI anomaly detection algorithms resort to unsupervised methods, assuming anomalies are hard to reconstruct or predict such as [10] and [11]. The disparity between the reconstructed or predicted data and the original data is utilized as the anomaly score and performs well under the assumption of Gaussian noise. Performance can falter with complex, non-Gaussian noise [12], especially in fluctuating time series data where differentiating anomalies is tough. Hence, some methods attempt to incorporate certain supervised information, as demonstrated by Huang et al. [13], who combine active learning and variational autoencoders to boost accuracy. However, these approaches necessitate a prior hypothesis of the data. A more universal and convenient choice would be a method capable of autonomously extracting features from the data itself.

Defining anomalies is intricate, often involving outliers with unusual data patterns [14] that necessitate contextual consideration [15]. For example, seasonal variations in power system operations can affect anomaly assessment. A comprehensive evaluation, including frequency domain insights, is crucial. Microsoft [4] employs spectral residual for processing one-dimensional KPIs, introducing frequency domain information to aid in anomaly identification. Park et al. [16] refine the Fast Fourier Transform (FFT) by computing only a subset of Fourier coefficients to expedite computation. The incorporation of frequency domain information contributes to enhancing detection effectiveness.

Additionally, monitoring data's multidimensionality introduces complex interdependencies among time and indicator dimensions [17]. Taking the IT operations domain as an example, as depicted in Figure 1, similar indicator types like CPU usage and load show related patterns, yet many methods incorrectly equate their correlation with dissimilar indicators such as disk and database metrics. Especially in anomaly detection, not all indicators exhibit anomalies simultaneously. Given the multitude of metrics in monitoring systems, treating various indicators equally may result in the features of indicators undergoing significant changes being submerged within numerous data points where no substantial changes have

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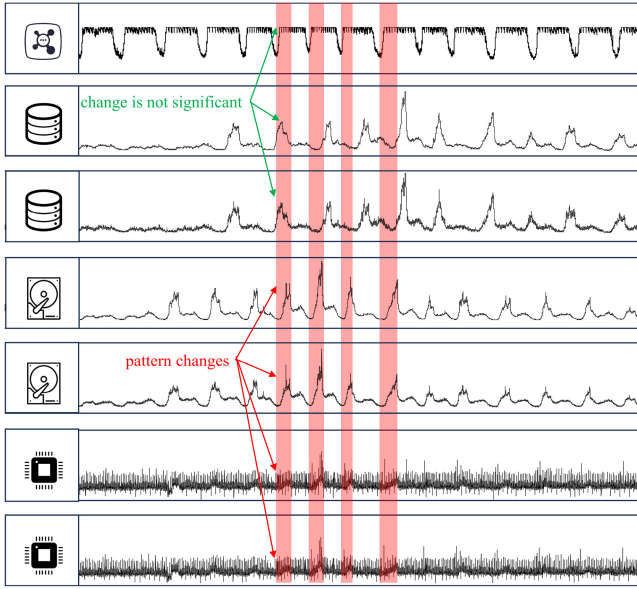


Fig. 1. Partial indicators obtained from an entity within the Internet monitoring system. The segments highlighted in red denote anomalies. Not all metrics are in an anomalous state when anomalies occur. The variables indicated by red arrows have experienced anomalies, while those indicated by green arrows have not exhibited significant changes. In particular, the Transformer structure facilitates extensive information exchange among different variables. Treating all variables equally may result in the changes occurring in anomalous indicators being averaged, potentially making it challenging to identify anomalous states comprehensively.

occurred, hindering timely anomaly capture. Exploring the correlations between variables and mitigating the interference of irrelevant variables on those experiencing anomalies can contribute to enhancing detection accuracy.

To address the challenges mentioned earlier, we propose the KPI anomaly detection method DMASA. First, the method tackles the issue of limited labels in KPI data by using self-supervised mask modeling. In contrast to contrastive learning, relying on prior assumptions and may introduce inductive bias, mask modeling excavates information from the data to learn richer representations with better generality. Secondly, it introduces frequency-domain information to identify anomalies from multiple perspectives based on spectral residual, which highlights anomalies by accentuating deviations from the frequency domain mean and enhances detection accuracy and timeliness. This helps the model in rapidly identifying the starting point of anomalies and capturing them earlier. Additionally, the distinct boundaries facilitate the observation of whether anomalies occur simultaneously across different variables.

Furthermore, treating all variables as equivalent may result in interference of irrelevant variables. Especially for models employing the Transformer method, akin to the “over-smoothing” issue [18], with the stacking of Transformer layers, variables tend to become increasingly similar, implying that local anomalous trends are flattened. Thus, DMASA introduces similarity aggregation to segregate irrelevant variables, preventing the dilution of anomaly signals and ensuring that only related variables influence the detection process. This targeted approach helps maintain the clarity of anomaly trends

across variables. Our main contributions can be distilled to the following points:

1. We introduce a novel anomaly detection approach that integrates mask modeling to adaptly capture the inherent structure of KPI data through self-supervised learning, overcoming the challenge of sparse labeling.
2. By utilizing spectral residual in conjunction with frequency-domain insights, our method bolsters the identification of anomalies, allowing for prompt detection and precise delineation of anomalous patterns.
3. Our model incorporates similarity aggregation to segregate irrelevant variables, ensuring that only closely related variables contribute to the reconstruction process. This refines the detection of anomalies without being obscured by unrelated data.
4. The vast majority of state-of-the-art experimental results obtained on eight datasets demonstrate the effectiveness and competitiveness of our proposed method.

The remaining sections of this paper are organized as follows. Section II provides a concise overview of KPI anomaly detection, the utilization of self-supervised methods in time series anomaly detection, and the application of Transformers in time series analysis. In Section III, we provide a detailed introduction to our DMASA method. Section IV comprehensively outlines the experimental details and analyzes the results of our approach. The article culminates with a conclusion that encapsulates our findings, contributions, and future works.

II. RELATED WORK

Operation and maintenance monitoring system analysis data comprise KPI data, logs, traces, and other relevant information, and KPI data draws widespread attention due to its capacity to effectively address dynamic changes in complex systems and facilitate prompt responses. First, we review self-supervised methods for unlabeled KPI data. Furthermore, recognizing the predominance of masking and predictive techniques within the Transformer framework in current self-supervised methods, we offer a comprehensive review of their application to time series data. Subsequently, we delve into the specifics of KPI anomaly detection, self-supervised methods for time series anomaly detection, and the utilization of Transformer-based strategies in time series analysis.

KPI Anomaly Detection: KPI data fundamentally is continuous time series data. Consequently, we will review methods designed for anomaly detection in time series data. They can be broadly categorized into statistical approaches, classical machine learning algorithms, and deep learning methods. Statistical methods, including mean-based approaches, feature prominently in this domain, with classic methods such as moving average [19] and Autoregression [20], autoregressive integrated moving average, and exponentially weighted moving average [21] being extensively explored. Classical machine learning algorithms, such as K-means [22], support vector machines [23], GBRT [24], and Gaussian Mixture Models [25], have also found applications. Deep learning methods encompass various network structures, including autoencoders [26], variational autoencoders (VAE) [27], temporal convolutional networks [28], recurrent neural

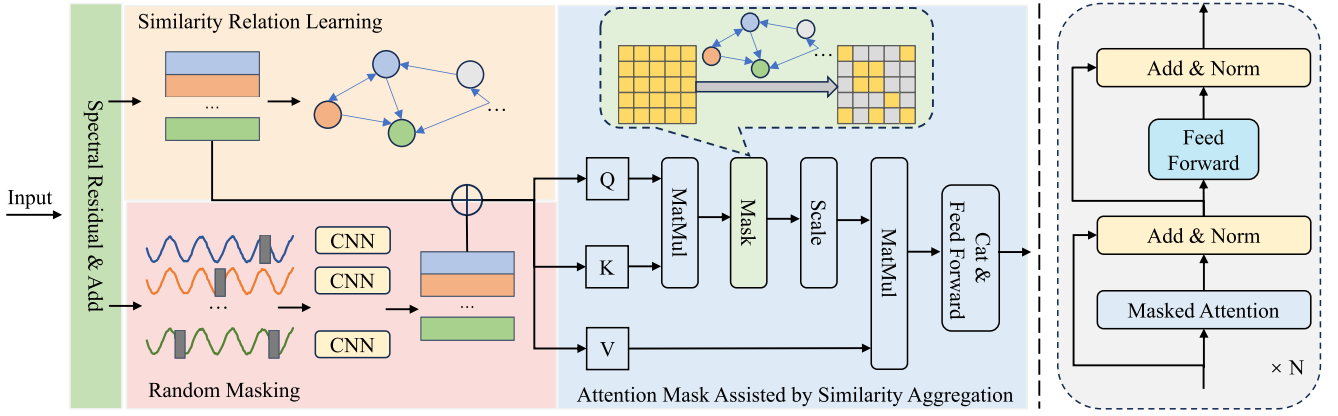


Fig. 2. DMASA network structure. Transformer Structure Combining Similarity Aggregation (Left): Following spectral residual processing, the unmasked sequence learns similarity relationships and embeddings. The masked sequence segment acquires hidden layer data through one-dimensional convolution. Both components are then integrated and fed into the Transformer block, where the attention mechanism incorporates the learned similarity relationships for information fusion. The sequence is ultimately reconstructed after this process.

networks (RNN) [29], [30], long short-term memory networks (LSTM) [31], graph neural networks [7], and Transformers. These methods operate under both supervised and unsupervised training paradigms. Supervised methods include EGADS [32] and Opprentice [33], while prevalent unsupervised methods include OmniAnomaly [29], GDN [7], TranAD [34], and Anomaly Transformer [35].

Self-Supervised Anomaly Detection Methods for Time Series: KPIs are predominantly characterized by unlabeled time series data. Self-supervised methods excel at distilling meaningful information from extensive data, thereby providing a robust representation for further analytical tasks. In the context of time series anomaly detection, self-supervised techniques predominantly harness contrastive learning strategies. For example, AnomalyBERT [36] defines and employs four types of methods to synthesize anomalous values, distinguishing positive and negative samples by replacing certain data with anomalous data. TimeAutoAd [37] uses three different strategies to augment training data for generating pseudo-negative time series and utilizes self-supervised contrastive loss to differentiate between original and generated time series. CSL [38] utilizes a shapelet-based encoder, incorporating multi-grained contrasting, multi-scale alignment, and a data augmentation library to learn time series features in a comprehensive manner. CTAD [39] incorporates seven types of general time-series data augmentation methods from the perspectives of variables and points, integrating a one-class learning approach into the contrastive learning loss. Despite the promising results of contrastive learning methods, they are not without limitations. The introduction of inductive bias based on researchers' prior assumptions can lead to performance inconsistencies when applied to time series data with different characteristics, as seen in the augmentation strategies employed by AnomalyBERT and TimeAutoAd. However, when transferred to time series problems in other domains with inconsistent data characteristics and prior assumptions, model performance can significantly decline.

Transformer Architecture for Time Series Analysis: Transformers have demonstrated remarkable success

in handling sequential data, including text and audio. Consequently, they have attracted considerable attention from researchers in the time series domain. Approaches like FEDformer [40] and PatchTST [41] have shown progress in time series prediction. In the realm of time series anomaly detection, Anomaly Transformer [35], AnomalyBERT [36], DCdetector [42], and others have achieved notable results. The application of Transformers in both prediction and anomaly detection signifies their versatility and effectiveness in handling diverse time series tasks.

III. APPROACH

A. Notation

KPI anomaly detection involves collecting metrics from monitored systems, with the aim of analyzing whether the system is in an abnormal state. Let d represent the number of variables and T_1 denote the length of the training data. We train on the given unlabeled KPI data $X'_{\text{train}} = (x_1, x_2, \dots, x_i, \dots, x_{T_1})$. For evaluation, we use the test set X_{test} for unsupervised assessment. We assess the model's performance by comparing its predictions with the test labels $Y_{\text{test}} = (y_1, y_2, \dots, y_i, \dots, y_{T_2})$, where $y_i \in \{0, 1\}$. Here, 0 indicates a normal state and 1 indicates an abnormal state.

B. Overall Framework

Our approach employs a self-supervised approach grounded in masked modeling to distill essential features from KPI data. Recognizing that KPI changes may not be synchronous across variables, our framework is designed to capture the inter-variable relationships, which is crucial for reducing the impact of irrelevant variables on the anomaly detection process. To achieve this, we integrate the learning of similarity relations and apply spectral residual methods to enhance the understanding of inter-variable dynamics and emphasize temporal changes.

As depicted in Figure 2, our model initiates with the spectral residual enhancement of the data, which is then merged with the original sequence. This processed data is directed into

two parallel components: Similarity Relation Learning and Random Masking. The Similarity Relation Learning module is tasked with identifying the relationships between variables, while the Random Masking module, following the application of random masks, is directed into a one-dimensional convolutional network to project the input data effectively. The outputs of these components are then consolidated and channeled into the Transformer module, where the similarity relation insights are deployed to modulate self-attention mechanisms. This strategic integration ensures that the model focuses on relevant connections between variables. This process is the reconstruction of the sequence, which is refined through minimizing reconstruction error, thereby optimizing the network's performance.

In the subsequent sections, we will delve deeper into the intricacies of each component and elucidate their synergistic contribution to the anomaly detection framework.

C. Enhancing Data With Spectral Residual

Anomalies, by nature, present a complex challenge to define and detect within data sets. In the context of KPI anomaly detection, while a certain degree of detection latency (the time lag between the anomaly's actual occurrence and its identification) might be tolerated, our goal is to minimize this delay to ensure the timeliness of monitoring systems. To boost the detection capabilities, we introduce the spectral residual method, which combines spectral information to amplify the deviation between each data point and the frequency-domain mean of its preceding data within a window. It is well-established that the frequency attributes of normal data differ markedly from those of anomalous data. When an anomaly emerges, the spectral residual method accentuates the divergence of the affected segment from the normal segment that precedes it, in terms of the frequency-domain mean. This amplification allows our network to more accurately and swiftly detect the onset of an anomaly, thereby reducing detection latency.

In multivariate scenarios, the spectral residual method further aids in identifying anomalies by accentuating their fluctuations, which in turn promotes the clustering of these points. This clustering facilitates the recognition of analogous patterns across variable fluctuations. Synchronous anomalies across variables suggest a heightened correlation between them, which is beneficial for our subsequent similarity aggregation processes. Our approach considers each of the d dimensions in the multivariate data as an independent univariate dataset, applying spectral residual operations to each one individually. By adding the spectral residual to the original data, we effectively increase the deviation of each data point, enhancing the model's sensitivity to anomalies. This methodical enhancement of the data not only refines the detection process but also enriches the model's ability to discern meaningful patterns within the data. The process is shown in Equation (1):

$$X_{\text{train}} = X'_{\text{train}} + SR(X'_{\text{train}}), \quad (1)$$

where the process of SR is described by Equations (2)-(5):

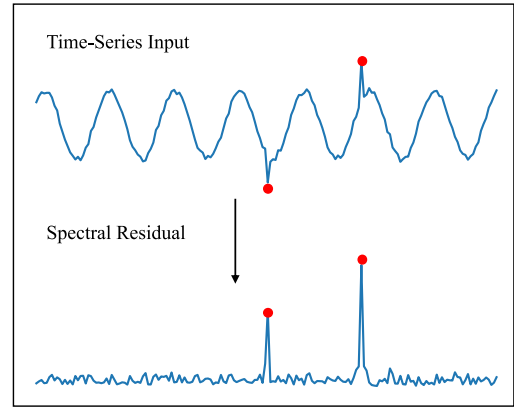


Fig. 3. Example of spectral residual results. The red points indicate anomalous data points. Following the application of spectral residual, the anomalous points are magnified.

$$A(f) = \text{Amplitude}(\mathfrak{F}(\mathbf{x})), \quad (2)$$

$$P(f) = \text{Phase}(\mathfrak{F}(\mathbf{x})), \quad (3)$$

$$R(f) = \log(A(f)) - I_p(f) \cdot \log(A(f)), \quad (4)$$

$$S(\mathbf{x}) = \left\| \mathfrak{F}^{-1}(\exp(R(f) + iP(f))) \right\|, \quad (5)$$

where $\mathfrak{F}$ and $\mathfrak{F}^{-1}$ denote Fourier Transform and Inverse Fourier Transform respectively. $\mathbf{x}$ is the input sequence of shape $n \times 1$. $A(f)$ is the amplitude spectrum of sequence $\mathbf{x}$. $P(f)$ represents the corresponding phase spectrum of the sequence. $I_p(f)$ is a local average filter of a $p \times p$ matrix, in which each element is $\frac{1}{p^2}$, to convolute the input sequence $\log(A(f))$.

Following spectral residual operations, the data exhibits the characteristics illustrated in the accompanying Figure 3. As observed, spectral residual amplifies the deviation of data from the frequency domain baseline. If a point is an anomaly, it will exhibit more pronounced behavior along the temporal dimension. Additionally, if multiple variables change simultaneously at a certain moment, it indicates a potentially closer relationship among them. The spectral residual method amplifies such changes, thereby distinguishing them from other variables that do not change synchronously, aiding in capturing the connections between variables.

D. Partitioning Variables Through Similarity Relationships

For KPI data, each dimension encapsulates a distinct metric or sensor, with each KPI exhibiting its own unique traits. The interplay among various features is often intricate. Consider a software operation system, where CPU-related metrics might fluctuate independently of database metrics. Similarly, in a water treatment system, the levels indicated by water level sensors may not consistently align with those of temperature sensors. In the realm of anomaly detection, it's crucial to recognize that anomalies across different variables are not uniformly synchronized. Applying a one-size-fits-all approach to variable analysis could potentially obscure the distinctive features of variables that exhibit anomalies, especially when other variables show no signs of anomalies. To counteract this, it's imperative to understand and leverage the underlying

dependencies between variables. This understanding allows for the aggregation of similar metrics, thereby minimizing the interference from irrelevant variables.

To achieve this, our methodology employs a learning process that discerns and respects the complex relationships between variables. By doing so, it ensures that the anomaly detection process is both sensitive and specific, acknowledging the unique characteristics of each KPI while identifying the subtleties of anomalies within their respective contexts. This approach is essential for developing a nuanced and effective anomaly detection system that can accurately interpret and respond to the multifaceted nature of KPI data.

Inspired by the graph neural network approach GDN [7], we employ a graph adjacency matrix to describe variable relationships. First, if we have prior knowledge about relationships between different metrics, we can incorporate this knowledge into the graph structure. Let A_{ij} represent the adjacency matrix indicating relationships between different variables, where $A_{ij} = 1$ indicates an edge from variable i to j , signifying an influence from variable i to j . However, in many cases, we may not have prior knowledge about the relationships between variables. We need to learn the proximity information from the data. Initially, we utilize $Embedding(\cdot)$ network to characterize each dimension of the variable x and get the embedding e_i ,

$$e_i \in \mathbb{R}^{d_{model}}, \text{ for } i \in \{1, 2, \dots, d\}. \quad (6)$$

Then, assess the proximity between variables by measuring cosine similarity,

$$v_{ji} = \frac{e_i^\top e_j}{\|e_i\| \cdot \|e_j\|}, \text{ for } j \neq i. \quad (7)$$

$$A_{ji} = 1, j \in Top - k(\{v_{ki} : k \in \{1, 2, \dots, d\} \setminus \{i\}\}). \quad (8)$$

We calculate the cosine similarity between embeddings for each pair of variables, and for each variable, we select the k variables corresponding to the embeddings with the highest similarity as neighbors. This indicates that these k neighbors are the variables most likely to influence it. The learning of embeddings e_i and the adjacency matrix A_{ij} will be utilized in subsequent information fusion using Transformer.

E. Self-Supervised Learning via Mask Modeling

KPI data, as a kind of time series data, exhibits temporal redundancy. As we claimed, some reconstruction methods assume that anomalies are challenging to identify based on reconstruction errors. However, the potent fitting capability of neural networks and the temporal redundancy in time series data make the reconstruction of anomalies less challenging. Conventional contrastive learning methods construct positive and negative samples and introduce inductive bias, limiting its universality and performance in various scenarios, as well as its transferability and generalization capabilities. To deal with it, we introduce mask modeling, starting from the distribution of the data itself, which achieves excellent results without additional assumptions about the data.

In this process, the primary steps of the model involve random masking and one-dimensional convolution processing

in the temporal dimension. Random masking involves independently selecting data in the time dimension with a proportion r for masking across different variable dimensions. That is, a proportion r of the data is masked in the time dimension, and variables at the same time may not necessarily be simultaneously masked across different variable dimensions. We take the window data $X = [x_t, x_{t+1}, \dots, x_{t+w-1}]$ as input. Let the mask matrix be denoted as M , which is a binary matrix with elements 0 and 1. The proportion of 0s in each row is r , and it is of the same dimension as X . After applying the mask, the updated representation X is as shown in the Equation (9):

$$X := M \odot X, \quad (9)$$

where $\odot$ represents element-wise multiplication. The one-dimensional convolution in the temporal dimension captures dependencies in the time series while also serving as a method for input projection. It is important to note that we process each variable separately, and the 1D-CNN is solely used to capture dependencies in a single dimension, as shown in the Equation (10),

$$X' = 1D - Conv(X). \quad (10)$$

F. Isolating Irrelevant Variables Through Similarity Aggregation

Transformer-based models excel in sequence tasks and are prevalent in sequence modeling. Yet, a common challenge arises with deep Transformers: as layers accumulate, information fusion can homogenize variables, obscuring anomalous trends. In cases where only a subset of system components malfunction, while others remain normal, the extensive interaction can mask the anomalies, leading to a false sense of normalcy and potential false negatives in monitoring.

To address this, we refine the Transformer architecture by limiting information fusion to closely related variables, thereby preserving the distinctiveness of anomalies. We integrate the Transformer with learned similarity relations to selectively fuse information, enhancing the model's ability to detect anomalies without being confounded by normal data variability. This nuanced approach ensures that only pertinent variables influence the anomaly detection process, improving the model's accuracy and reliability.

In comparison to a standard Transformer, our enhancements are primarily focused on the self-attention mechanism. Here, assuming $E = [e_1, e_2, \dots, e_i, \dots, e_d]$, we use a linear transformation $W_3 X'$ for value V , $E \oplus W_1 X'$ and $E \oplus W_2 X'$ as query Q and key K for self-attention calculation. Note that the components at corresponding positions of E and $W_1 X'$ represent the same variables. Moreover, attention needs to incorporate the adjacency matrix A , with the attention matrix result being valid only when $A_{ij} = 1$. As shown in the following Equation (11),

$$Z = softmax(AQK^T)V. \quad (11)$$

Algorithm 1 The Training Phase of DMSA

Input: $X'_{train} \in \mathbb{R}^{d \times T_1}$, window size w of X'_{train} ;
Output: Network that maps to the embedding $Embedding(\cdot)$, Adjacency matrix A , $1D-Conv(\cdot)$, Fully connected network f_θ , Our Transformer network incorporating the masking mechanism $ModifiedTransformer(\cdot)$;
 $X'_{train} \leftarrow X'_{train} + SR(X'_{train})$;
 $N_{train} \leftarrow \frac{T_1}{w}$;
for $n_{train} \leftarrow 1$ **to** N_{train} **do**
 $X \leftarrow X'_{train}[n_{train} \cdot w:(n_{train} + 1) \cdot w]$;
for $i \leftarrow 1$ **to** d **do**
 $e_i \leftarrow Embedding(X_{i*})$;
end for
for $i \leftarrow 1$ **to** d **do**
for $j \leftarrow 1$ **to** d **do**
 $v_{ji} = \frac{e_i^\top e_j}{\|e_i\| \cdot \|e_j\|}, j \neq i$;
end for
 $A_{ji} \leftarrow 1$ **if** j **in** $Top-k(v_{ki}, k \neq i)$;
end for
 $M \in \mathbb{R}^{D \times w}$, with $M_{ij} \in \{0, 1\}$;
 $X' = ID-Conv(M \odot X)$;
 $E = [e_1, e_2, \dots, e_i, \dots, e_d]$;
 $Z_{hidden} = ModifiedTransformer(E, X', A)$;
 $Loss_{MAE} = \frac{1}{m} \|f_\theta(Z_{hidden}) - X\|$;
end for
utilize $Loss_{MAE}$ to improve the entire network;

Additionally, before entering the feedforward network, we perform a dot product between embeddings E and Z , as shown in the following Equation (12),

$$Z := EZ. \quad (12)$$

After several Transformer layers, we utilize a simple fully connected network to aggregate data from different dimensions. The Mean Absolute Error (MAE) reconstruction error is employed as the optimization objective, aiming to minimize the overall objective function. As shown in the following Equation (13), (14),

$$Z' = f_\theta(Z_{hidden}), \quad (13)$$

$$Loss_{MAE} = \frac{1}{m} \|Z' - X\|, \quad (14)$$

where f_θ is a fully connected network, and m is the number of matrix elements. Z_{hidden} represents the output data obtained after the input data has undergone the modified Transformer component. The comprehensive algorithm for the training phase is illustrated in Algorithm 1.

G. Anomaly Detection Scoring

Finally, in the process of assigning anomaly scores, we enhance the evaluation using spectral residual. Spectral residual amplifies the deviation of values from a frequency domain perspective. By leveraging the disparity between the data processed through spectral residual and the original data, we

Algorithm 2 The Testing Phase of DMSA

Input: $X_{test} \in \mathbb{R}^{d \times T_2}$, $Embedding(\cdot)$, A , $1D-Conv(\cdot)$, $ModifiedTransformer(\cdot)$, f_θ , w ;
Output: $AnomalyScore$
 $N_{test} \leftarrow \frac{T_2}{w}$;
for $n_{test} \leftarrow 1$ **to** N_{test} **do**
 $X \leftarrow X_{test}[n_{test} \cdot w:(n_{test} + 1) \cdot w]$;
for $i \leftarrow 1$ **to** d **do**
 $e_i \leftarrow Embedding(X_{i*})$;
end for
 $X' = ID-Conv(X)$;
 $E = [e_1, e_2, \dots, e_i, \dots, e_d]$;
 $Z_{hidden} = ModifiedTransformer(E, X', A)$;
 $Z' = f_\theta(Z_{hidden})$;
 $AnomalyScore = \frac{1}{m} \|Z' - X\| + SR(\frac{1}{m} \|Z' - X\|)$;
end for

assess whether the current time point is in an anomalous state based on the level of deviation:

$$AnomalyScore = \frac{1}{m} \|Z' - X\| + SR(\frac{1}{m} \|Z' - X\|) \quad (15)$$

The comprehensive algorithm for the testing phase is illustrated in Algorithm 2. In line with standard procedures in current self-supervised studies, our testing phase begins with scoring each test sample, followed by threshold adjustment to align with the benchmark's anomaly likelihood. Anomalies are flagged when their scores exceed this threshold. This process mirrors the operations within real-world systems. A better detection model can offer clearer statistical distinctions between normal data points and anomalies, facilitating the determination of an appropriate threshold to perform this task more effectively.

H. Computational Complexity

Our proposed method incurs two additional computational costs compared to conventional Transformer approaches. Firstly, we employ spectral residuals to augment the data, which involves the use of FFT with a time complexity of $O(d \cdot T_1 \log T_1)$. Secondly, the method includes learning similarity relationships, which entails calculating pairwise neighbor relationships between embeddings of dimension d_{model} across d variable dimensions, resulting in a complexity of $O(d \cdot d_{model}^2)$. As a comparison, the Anomaly Transformer requires additional Gaussian kernel overhead, while the DCdetector applies self-attention both within and between patches to facilitate contrastive learning. In comparison, our method does not result in a significant increase in overhead. With the GPU acceleration, the additional computational overhead introduced by our method is acceptable.

IV. EXPERIMENTS

A. Benchmark Dataset

We employ five of the most classic and widely compared datasets, including PSM (Pooled Server Metrics), MSL (Mars Science Laboratory rover), SMAP (Soil Moisture

Active Passive satellite), SWaT (Secure Water Treatment), and SMD (Server Machine Dataset). Additionally, to visually demonstrate the model's detection capabilities across various anomaly types, we utilize NeurIPS-TS. Furthermore, we add two real-world multidimensional time series datasets from different domains: NIPS-TS-SWAN and NIPS-TS-GECCO. Here is a brief overview of the datasets, as shown in Table II:

1) *PSM*: The PSM dataset, featuring 25 dimensions, is collected from multiple application server nodes at eBay. It includes 13 weeks of training data and 8 weeks of testing data, and is characterized by a relatively high anomaly ratio.

2) *MSL & SMAP*: The MSL dataset contains sensor data from NASA's Mars Curiosity Rover, while the SMAP dataset, comprising 25 dimensions, includes soil samples and telemetry from the Soil Moisture Active Passive (SMAP) satellite. The data represents real spacecraft telemetry and anomalies from both missions, fully anonymized with respect to channel IDs.

3) *SWaT*: The dataset, consisting of 51 dimensions, is collected from a continuously operating water treatment infrastructure. It spans 11 days of network traffic and sensor data, with 7 days of normal operation and 4 days of attack scenarios involving 41 documented attacks, all labeled as either normal or abnormal behavior.

4) *SMD*: A five-week dataset gathered from an Internet company's cluster. As per the instructions provided by the data provider, the detection process is required to be carried out on individual entities separately. Thus, we chose the first entity from the first cluster.

5) *NeurIPS-TS*: A synthetic dataset provided by [45] for detecting various anomaly types.

6) *NIPS-TS-SWAN*: The NIPS-TS-SWAN dataset, focused on drinking water quality in the "Internet of Things" context, featuring a relatively high anomaly rate and posing challenge for anomaly detection.

7) *NIPS-TS-GECCO*: The NIPS-TS-GECCO dataset is derived from vector magnetograms of the solar photosphere in Spaceweather HMI active regions, and presents a very low anomaly rate, making it another difficult anomaly detection dataset.

B. Baselines

We compared our model with 16 different approaches for a comprehensive evaluation. DAGMM [25], LOF [48], MPPCAD [49] utilize density estimation models; OC-SVM [23], iForest [50] employ classical machine learning algorithms. Deep SVDD [51], THOC [52], ITAD [53] use clustering methods, VAR [54], LSTM [31] are autoregressive models, and LSTM-VAE [26], OmniAnomaly [29], BeatGAN [55], InterFusion [56], Anomaly Transformer [35], DCdetector [42] are reconstruction-based models. Among reconstruction-based methods, LSTM-VAE employs LSTM to extract temporal features and reconstruct the expected distribution of the data by progress-based varying prior. OmniAnomaly employs stochastic variable connection and planar normalizing flow to acquire robust representations of time series data. BeatGAN utilizes an adversarial generative approach coupled with time series warping for reconstructing

sequences. InterFusion employs a hierarchical VAE to learn low-dimensional inter-metric and temporal embeddings. Anomaly Transformer detects anomalies by leveraging differences between local and global correlations in normal and anomalous data. DCdetector combines contrastive learning to learn permutation invariant representations from a multi-scale perspective.

Common evaluation metrics, including accuracy, precision, recall, and F1-score, are utilized. In alignment with [11], [29], [35], [42], [52], we adopt common adjustment strategies for a fair evaluation. Additionally, we have supplemented recently proposed metrics such as affiliation precision/recall [57], Range-AUC-ROC, Range-AUC-PR, V_ROC, and V_PR [58]. Affiliation precision and recall [57] are determined by measuring the distance between true labels and predicted labels. Range AUC-ROC and Range AUC-PR add a buffer transition area at anomaly boundaries, mapping discrete data to a continuous region and calculating the area under the ROC and PR curves for different thresholds. The VUS metric represents the volume corresponding to Range AUC-ROC and Range AUC-PR for various buffer sizes [58].

C. Implementation Details

We summarize the default hyper-parameters as follows. Our model has a default of 3 layers for the Transformer. Except for the SMAP dataset, where the hidden layer dimension is set to 64, the default dimension for hidden layers in other datasets is 256. The default stride for one-dimensional convolution is set to 32. For convenience in masking, the number of heads in attention is set to 1 by default. The default masking rate for the mask part is 0.3, the number of nearby variables (Top-k) is set to 3, the window size is 128, and the default definition for the anomaly threshold is 1. We utilize the Adam optimizer to optimize network parameters, setting the initial learning rate to 10^{-3} . The default batch size for both training and testing is 128. All experiments are conducted using the PyTorch 1.7.1 library on a computer equipped with a GeForce RTX 3060 GPU supporting NVIDIA CUDA.

D. Research Question

In the following experiments, we endeavored to answer these questions.

•*RQ1*: Faced with the challenge of unlabeled KPI data, has the overall design of the DMSA framework resulted in effective anomaly detection?

•*RQ2*: Do the various components of DMSA collaborate effectively to identify different types of anomalies, including point anomalies and pattern anomalies?

•*RQ3*: Has the use of similarity aggregation to isolate irrelevant variables and the application of spectral residual to strengthen anomaly boundaries yielded positive results?

•*RQ4*: Is similarity aggregation meaningful? Do features of different variables exhibit meaningful correlations, either proximity or divergence?

•*RQ5*: How does the model's performance vary under different parameters? Is it robust?

TABLE I

RESULTS ON FIVE WIDELY USED REAL-WORLD DATASETS FOR PERFORMANCE COMPARISON. PERFORMANCE IS RANKED IN ASCENDING ORDER BASED ON F1 SCORE. P, R, AND F1 REPRESENT PRECISION, RECALL, AND F1 SCORE, RESPECTIVELY. ALL RESULTS ARE PRESENTED IN PERCENTAGES (%), WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND-BEST RESULTS INDICATED BY A BLUE UNDERScore. THE RESULTS OF OTHER METHODS ARE SOURCED FROM DCDETECTOR [42]

Method	Dataset	SMD			MSL			SMAP			SWaT			PSM		
	Metric	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Classic	OCSVM	44.34	76.72	56.19	59.78	86.87	70.82	53.85	59.07	56.34	45.39	49.22	47.23	62.75	80.89	70.67
	iForest (ICDM 2008)	42.31	73.29	53.64	53.94	86.54	66.45	52.39	59.07	55.53	49.29	44.95	47.02	76.09	92.45	83.48
Clustering	Deep-SVDD (ICML 2018)	78.54	79.67	79.10	91.92	76.63	83.58	89.93	56.02	69.04	80.42	84.45	82.39	95.41	86.49	90.73
	THOC (NeurIPS 2020)	79.76	90.95	84.99	88.45	90.97	89.69	92.06	89.34	90.68	83.94	86.36	85.13	88.14	90.99	89.54
	ITAD (CIKM 2020)	86.22	73.71	79.48	69.44	84.09	76.07	82.42	66.89	73.85	63.13	52.08	57.08	72.80	64.02	68.13
Density	LOF (SIGMOD 2000)	56.34	39.86	46.68	47.72	85.25	61.78	58.93	56.33	57.60	72.15	65.43	68.62	57.89	90.49	70.61
	MMPCACD (TAES 2017)	71.20	79.28	75.02	81.42	61.31	69.95	88.61	75.84	81.73	82.52	68.29	74.73	76.26	78.35	77.29
	DAGMM (ICLR 2018)	67.30	49.89	57.30	89.60	63.93	74.62	86.45	56.73	68.51	89.92	57.84	70.40	93.49	70.03	80.08
Autoregression	VAR	78.35	70.26	74.08	74.68	81.42	77.90	81.38	53.88	64.83	81.59	60.29	69.34	90.71	83.82	87.13
	LSTM (KDD 2018)	78.55	85.28	81.78	85.45	82.50	83.95	89.41	78.13	83.39	86.15	83.27	84.69	76.93	89.64	82.80
Reconstruction	LSTM-VAE (RAL 2018)	75.76	90.08	82.30	85.49	79.94	82.62	92.20	67.75	78.10	76.00	89.50	82.20	73.62	89.92	80.96
	OmniAnomaly (KDD 2019)	83.68	86.82	85.22	89.02	86.37	87.67	92.49	81.99	86.92	81.42	84.30	82.83	88.39	74.46	80.83
	BeatGAN (ICAI 2019)	72.90	84.09	78.10	89.75	85.42	87.53	92.38	55.85	69.61	64.01	87.46	73.92	90.30	93.84	92.04
	InterFusion (KDD 2021)	87.02	85.43	86.22	81.28	92.70	86.62	89.77	88.52	89.14	80.59	85.58	83.01	83.61	83.45	83.52
	AnomalyTrans (ICLR 2022)	<u>88.47</u>	<u>92.28</u>	<u>90.33</u>	91.92	<u>96.03</u>	<u>93.93</u>	93.59	99.41	96.41	89.10	99.28	94.22	96.94	97.81	97.37
	DCdetector (KDD 2023)	83.59	91.10	87.18	93.69	99.69	96.60	<u>95.63</u>	<u>98.92</u>	<u>97.02</u>	<u>93.11</u>	99.77	<u>96.33</u>	<u>97.14</u>	98.74	<u>97.94</u>
Ours		96.11	100.00	98.02	<u>92.06</u>	95.45	93.73	98.33	96.00	97.15	95.82	<u>99.31</u>	97.53	98.09	<u>98.25</u>	98.17

TABLE II

BENCHMARK DATASET DETAILS, WHERE DIM REPRESENTS DIMENSIONS, AND AR REPRESENTS ANOMALY RATIO

Benchmarks	Source	Dim	#Train	#Test	AR(%)
PSM [43]	Server	25	132,481	87,841	27.8
MSL [31]	Space	55	58,317	73,729	10.5
SMAP [31]	Space	25	135,183	427,617	12.8
SWaT [44]	Water	51	495,000	449,919	12.1
SMD [29]	Server	38	28,479	28,479	9.5
NeurIPS-TS [45]	Synthesis	1	10,000	10,000	1.0
NIPS-TS-SWAN [46]	Weather	38	60,000	60,000	32.6
NIPS-TS-GECCO [47]	Water	9	69,260	69,261	1.1

E. Main Results

RQ1: We initially evaluate our model on five widely-used and well-assessed real-world scenario datasets, comparing it against 16 various baseline models as presented in Table I. Our model continues to exhibit superiority compared to advanced models like Anomaly Transformer [35] and DCdetector [42]. Notably, using the same evaluation metrics as the Anomaly Transformer and DCdetector, our method achieves state-of-the-art performance on most datasets. Despite the impressive performance of current methods—DCdetector, for instance, has garnered F1 scores of 96.33% on SWaT and 97.94% on PSM benchmarks—our approach has managed to outperform them, achieving F1 scores of 97.53% on SWaT and 98.17% on PSM. We eschew the use of contrastive learning strategies for sample augmentation, which, while beneficial for some methods like DCdetector in enhancing performance on the MSL benchmark, incurs higher training costs and can compromise generalization across other benchmarks, as evidenced by the 87.18% F1 score on the SMD benchmark. In comparison, our method not only demonstrates superior performance across all benchmarks but also significantly elevates the overall detection capabilities.

Despite ongoing controversies in evaluating anomaly detection algorithms, we argue that timely alerts and interventions during anomalies align with the ultimate goal of anomaly

detection in practice. Therefore, we adhere to the evaluation methods consistent with the Anomaly Transformer and DCdetector. Of course, for a more comprehensive assessment of the performance of various methods, we complement the additional evaluation with recently proposed metrics such as affiliation precision/recall [57], Range-AUC-ROC, Range-AUC-PR, V_ROC, and V_PR [58]. Table I illustrates that the best results are concentrated among Anomaly Transformer, DCdetector, and our method. Therefore, we provide a broader comparison of these three methods in Table III. The table demonstrates that our method excels on most metrics and datasets. The choice of anomaly threshold also affects the corresponding evaluation metrics. If the goal is accurate anomaly detection, aligning with our method involves selecting a smaller anomaly rate to raise the anomaly threshold. Conversely, if the focus in anomaly detection is on the distance between real situations and predicted events, selecting a larger anomaly rate to relax the threshold produces better results.

Furthermore, considering the already high evaluation results on several commonly used datasets, we increased the difficulty of anomaly detection by employing more challenging datasets for comparison among various anomaly detection methods. We supplemented experimental data for the NIPS-TS-SWAN and NIPS-TS-GECCO datasets in Table VI. These datasets present challenging detection scenarios with anomaly rates of 32.6% and 1.1%, respectively. We compared our model with several representative methods that have demonstrated good performance, including Matrix Profile, GBRT, LSTM-RNN, Autoregression, OCSVM, AutoEncoder, Anomaly Transformer, iForest, and DCdetector. Remarkably, our model still achieved the best overall performance, underscoring the competitiveness of our approach. Similarly, considering that outstanding deep learning methods are still Anomaly Transformer and DCdetector, we further compared our model with these two methods. The results across different metrics for the three models are presented in Table IV, comprehensively showcasing the performance of our model. In terms of both detection accuracy and the rapidity of response,

TABLE III

ADDITIONAL EVALUATION METRICS ON THE 5 DATASETS. AFF-P REPRESENTS AFFILIATION PRECISION, AFF-R STANDS FOR AFFILIATION RECALL [57], R_A_R, AND R_A_P [58] ARE RANGE-AUC-ROC AND RANGE-AUC-PR, RESPECTIVELY, REPRESENTING TWO SCORES BASED ON LABEL TRANSFORMATION UNDER ROC AND PR CURVES. V_ROC AND V_PR [58] REPRESENT VOLUMES UNDER THE SURFACE CREATED BY ROC AND PR CURVES, RESPECTIVELY. ALL RESULTS ARE PRESENTED IN PERCENTAGES (%), WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND-BEST RESULTS INDICATED BY A BLUE UNDERScore. THE TWO VALUES OF AR REPRESENT THE EVALUATION RESULTS OF OUR MODEL AT DIFFERENT THRESHOLDS

Dataset	Method	Acc	P	R	F1	Aff-P	Aff-R	R_A_R	R_A_P	V_ROC	V_PR
SWaT	AnomalyTrans*	98.51	89.10	99.28	94.22	53.03	98.08	<u>97.89</u>	93.47	<u>97.92</u>	93.49
	DCdetector*	<u>99.09</u>	93.11	99.77	96.33	52.40	<u>97.67</u>	96.63	<u>94.06</u>	96.95	<u>94.34</u>
	Ours(AR=0.6)	99.39	95.82	99.31	97.53	56.66	94.96	95.18	93.99	95.42	94.20
	Ours(AR=1.1)	<u>99.09</u>	<u>93.37</u>	<u>99.59</u>	<u>96.38</u>	<u>53.99</u>	97.11	98.08	95.47	98.07	95.45
PSM	AnomalyTrans*	98.68	96.94	97.81	97.37	55.35	80.28	<u>91.83</u>	<u>93.03</u>	<u>88.71</u>	<u>90.71</u>
	DCdetector*	<u>98.95</u>	<u>97.14</u>	98.74	<u>97.94</u>	54.71	<u>82.93</u>	91.55	92.93	88.41	90.58
	Ours(AR=1.1)	98.98	98.09	98.25	98.17	62.20	76.78	88.32	91.30	84.55	88.57
	Ours(AR=2.0)	98.22	95.35	<u>98.39</u>	96.85	<u>58.68</u>	84.73	93.68	94.18	89.95	91.50
SMAP	AnomalyTrans*	99.05	93.59	99.41	96.41	51.39	98.68	96.32	<u>94.07</u>	95.52	<u>93.37</u>
	DCdetector*	<u>99.21</u>	<u>95.63</u>	<u>98.92</u>	<u>97.02</u>	<u>51.46</u>	<u>98.64</u>	<u>96.03</u>	94.18	<u>95.19</u>	93.46
	Ours(AR=0.3)	99.28	98.33	96.00	97.15	55.89	82.19	78.66	80.82	78.65	80.80
	Ours(AR=2.1)	98.09	87.99	98.51	92.95	49.80	93.29	90.41	86.65	90.24	86.50
MSL	AnomalyTrans*	<u>98.69</u>	91.92	96.03	<u>93.93</u>	51.76	95.98	<u>90.04</u>	<u>87.87</u>	88.20	<u>86.26</u>
	DCdetector*	99.06	93.69	99.69	96.60	<u>51.84</u>	<u>97.39</u>	93.17	91.64	93.15	91.66
	Ours(AR=1.0)	98.65	<u>92.06</u>	95.45	93.73	51.72	93.88	85.35	83.84	85.43	83.93
	Ours(AR=2.0)	97.96	85.19	<u>97.59</u>	90.97	52.93	97.68	89.29	84.51	<u>89.13</u>	84.37

* indicates the results which are adopted from DCdetector [42]

TABLE IV

COMPARISON OF ADDITIONAL METRICS FOR THE NIPS-TS DATASET. ALL RESULTS ARE PRESENTED IN PERCENTAGES (%), WITH THE BEST RESULTS HIGHLIGHTED IN RED. THE RESULTS OF OTHER METHODS ARE SOURCED FROM DCDETECTOR [42]

Dataset	Method	Acc	P	R	F1	Aff-P	Aff-R	R_A_R	R_A_P	V_ROC	V_PR
NIPS-TS-SWAN	AnomalyTrans	84.57	90.71	47.43	62.29	58.45	9.49	86.42	93.26	84.81	92.00
	DCdetector	85.94	95.48	59.55	73.35	50.48	5.63	88.06	94.71	86.25	93.50
	Ours	86.20	96.44	59.74	73.78	55.49	5.25	88.08	94.79	83.45	91.89
NIPS-TS-GECCO	AnomalyTrans	98.03	25.65	28.48	26.99	49.23	81.20	56.35	22.53	55.45	21.71
	DCdetector	98.56	38.25	59.73	46.63	50.05	88.55	62.95	34.17	62.41	33.67
	Ours	98.92	49.08	62.05	54.80	58.25	88.82	62.36	39.06	63.55	39.48

indicating the distance between real situations and predicted events, our method consistently maintains a leading position.

F. Visualization of Anomaly Score

RQ2: To demonstrate the effectiveness of our model in handling various types of anomalies, we adhere to the benchmark conventions outlined in NeurIPS-TS and visually present anomaly scores as shown in Figure 4. The NeurIPS-TS benchmark is a meticulously designed artificially synthesized anomaly detection benchmark that encompasses point anomalies, including global point anomalies and contextual point anomalies, as well as pattern anomalies such as shapelet anomalies, seasonal anomalies, and trend anomalies. In particular, from the experimental results, it can be observed that our model is capable of promptly identifying and pinpointing point anomalies. Moreover, for pattern anomalies,

our model efficiently recognizes the onset points where patterns undergo abrupt changes, rapidly localizing the start and end points of anomalous segments. Clearly, our model proves capable of effectively addressing different types of anomalies.

G. Ablation Studies

RQ3: Table V presents the ablation study of the two main modules under consistent parameters. The attention mask designates a module designed to discern similarity relations and impose similarity adjacency structure onto the attention module of the Transformer. This module restricts the fusion of information across distinct variable dimensions, as elaborated upon in Section III-D and Section III-F concerning the topic of similarity aggregation. Spectral residual, on the other hand, is the part of Section III-C and III-G that utilizes spectral

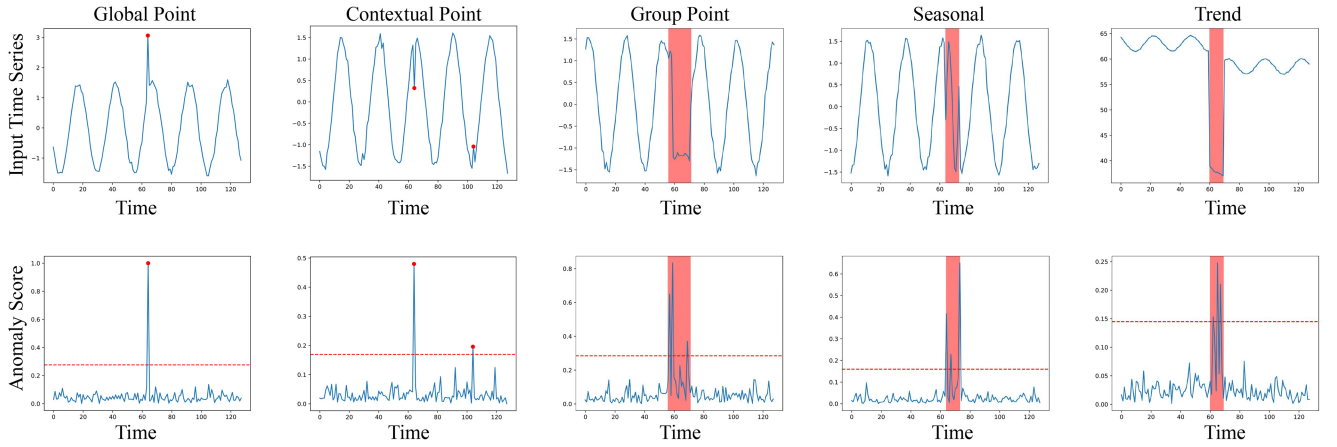


Fig. 4. Visualization of the detection of different types of anomalies by our model. It can be observed that for point anomalies, DMASA yields significantly higher anomaly scores, allowing for the easy identification of anomalous points. For pattern anomalies, DMASA demonstrates the ability to locate such anomalies by exhibiting notably elevated anomaly scores at the starting and ending points of the anomalous segment.

TABLE V

ABLATION STUDY OF THE MODEL, WHERE ATTENTION MASK REPRESENTS THE PART RELATED TO SIMILARITY AGGREGATION, AND SPECTRAL RESIDUAL REPRESENTS THE PART RELATED TO SPECTRAL RESIDUAL. ALL RESULTS ARE PRESENTED IN PERCENTAGE (%), WITH THE BEST RESULTS HIGHLIGHTED IN RED

Attention Mask	Spectral Residual	MSL			SMAP			SWaT			PSM			SMD		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
✗	✓	91.45	89.69	90.56	96.79	92.48	94.58	96.64	96.77	96.71	97.77	95.22	96.48	91.39	99.74	95.39
✓	✗	91.62	91.28	91.45	96.45	57.04	71.69	95.73	98.58	97.14	97.28	96.12	96.70	91.79	100.00	95.72
✓	✓	92.06	95.45	93.73	98.33	96.00	97.15	95.82	99.31	97.53	98.09	98.25	98.17	96.11	100.00	98.02

TABLE VI

RESULTS FOR THE NIPS-TS DATASET. PERFORMANCE IS RANKED FROM LOW TO HIGH. ALL RESULTS ARE PRESENTED IN PERCENTAGES (%), WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND-BEST RESULTS INDICATED BY A BLUE UNDERScore. THE RESULTS OF OTHER METHOD SOURCED FROM DCDETECTOR [42]

Dataset	NIPS-TS-GECCO			NIPS-TS-SWAN		
Metric	P	R	F1	P	R	F1
MatrixProfile	4.6	18.5	7.4	16.7	17.5	17.1
GBRT	17.5	14.0	15.6	44.7	37.5	40.8
LSTM-RNN	34.3	27.5	30.5	52.7	22.1	31.2
Autoregression	39.2	31.4	34.9	42.1	35.4	38.5
OCSVM	18.5	74.3	29.6	47.4	49.8	48.5
AutoEncoder	42.4	34.0	37.7	49.7	52.2	50.9
AnomalyTrans	25.7	28.5	27.0	90.7	47.4	62.3
iForest	<u>43.9</u>	35.3	39.1	56.9	59.8	58.3
DCdetector	38.3	59.7	<u>46.6</u>	<u>95.5</u>	59.6	<u>73.4</u>
Ours	49.1	<u>62.1</u>	54.8	96.4	<u>59.7</u>	73.8

residuals to enhance the data, combining frequency domain data to aid in learning similarity relations and reinforcing the scoring mechanism. It can be observed that, although the performance improvement seems not very significant on some datasets such as MSL and PSM, there is a substantial enhancement, particularly with the spectral residual method on the SMAP dataset. The reason for this could be the high anomaly rate in the SMAP dataset, along with a much larger size of the test set compared to the training set, resulting in overall large-scale and challenging detection scenarios. The spectral residual method provides significant enhancement

from a frequency domain perspective, leading to substantial performance improvement on the SMAP dataset.

H. Visualization of Variable Relationships

RQ4: In order to provide a more comprehensive and visually intuitive depiction of the relationships between variables, we computed similarity using the e_i embeddings learned from the PSM dataset and represented them through a heatmap, as illustrated in Figure 6. The color intensity in the heatmap corresponds to the strength of the correlation between variables, with darker colors indicating stronger correlations. Evidently, for each variable, only a limited set of other variables exhibit high correlation. This observation suggests that when employing attention mechanisms for information fusion among variables, focusing on these highly correlated variables is sufficient. This focused attention facilitates the effective transmission of relevant information, reducing the impact of irrelevant variables on anomaly detection. Simultaneously, this strategy contributes to a reduction in computational complexity during self-attention computations.

I. Parameter Sensitivity

RQ5: Simultaneously, we investigated the sensitivity of our model to different parameters, as illustrated in the Figure 5. The *mask_ratio* represents the difficulty of reconstructing data for the model. It is noteworthy that even with a high masking rate, our model maintains robust detection performance. This robustness may be attributed, on one hand, to the residual connection structure of the Transformer, allowing

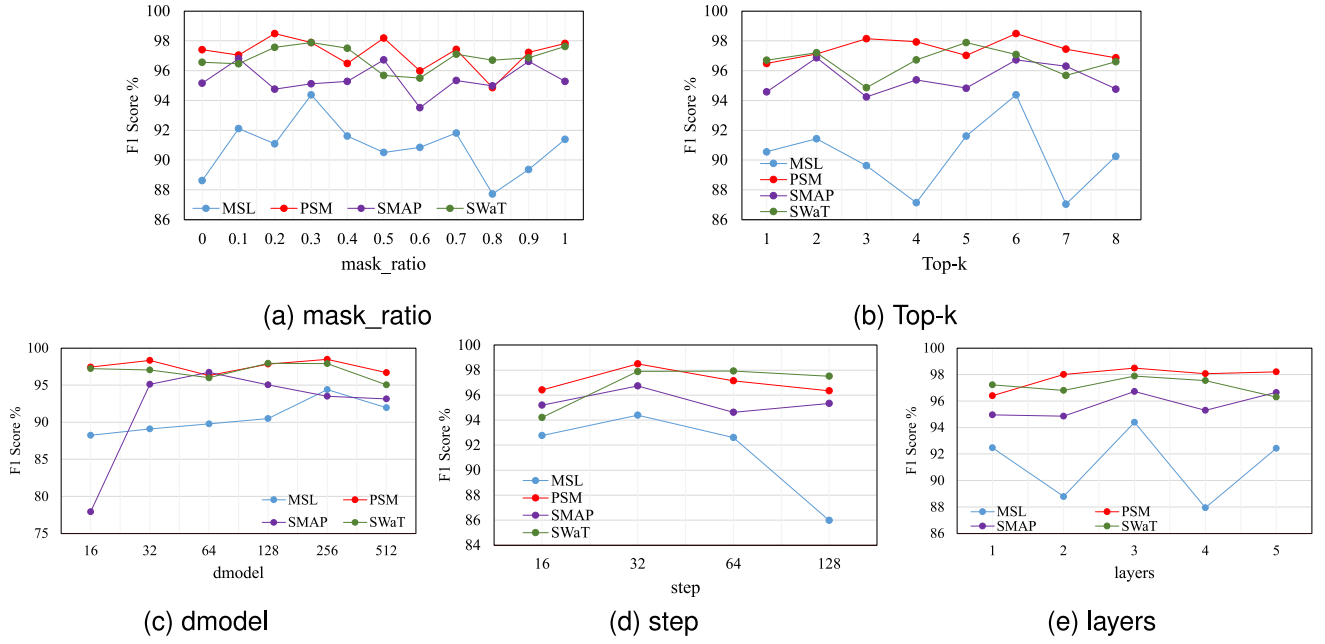


Fig. 5. Sensitivity analysis of key parameters. *mask_ratio* denotes the proportion of masked data during self-supervised learning. *Top-k* represents the number of other variables involved in information fusion for each variable. *dmodel* signifies the dimensionality of the hidden layer variables. *step* denotes the stride in one-dimensional convolution. *layers* indicate the number of layers in the stacked Transformer.

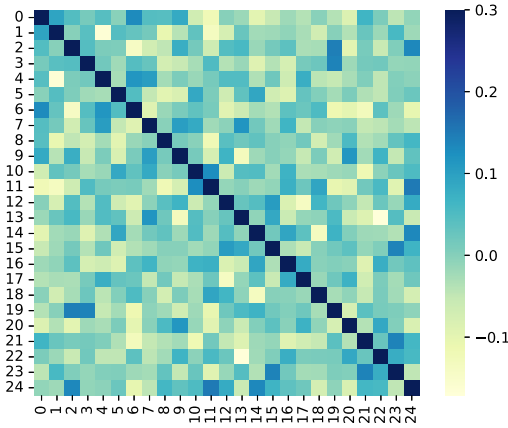


Fig. 6. Visualization of pairwise similarity among different variables. The darker the color, the greater the degree of correlation between the two variables.

information to be effectively transmitted. On the other hand, it could be because the learned embeddings e_i carry essential information that remains unaffected by masking. The parameter *dmodel* corresponds to the dimensionality of the intermediate variable, and *layers* denotes the number of layers in the Transformer, both of which can influence computational efficiency. Increasing *dmodel* and *layers* sometimes leads to improved detection efficiency. However, considering the associated performance overhead, we opt for a trade-off. The parameter *step* represents the stride during one-dimensional convolution, indicating the aggregation of several time points. *Top-k* determines the number of most relevant variables selected, influencing the number of related variables each variable can “see”. It is observed that allowing each variable to interact with all variables may introduce interference, leading to a decrease in detection efficiency. Therefore, selecting a

few most relevant variables for information fusion is sufficient. In the case of the MSL benchmark, the lack of a contrastive strategy to enrich the data, along with a smaller training dataset relative to the test dataset, leads to a slightly lower robustness. However, the overall resilience of our method is confirmed by its performance on other benchmarks. Balancing training expenses with the performance trends shown in Figure 5 (b), we recommend setting the parameter *k* optimally between 5 and 6.

V. CONCLUSION AND FUTURE WORK

KPI anomaly detection plays a crucial role in maintaining the overall reliability of operational systems. However, due to the lack of labels in KPI data, self-supervised approaches are proposed to learn data features for anomaly detection. In this work, we introduce a pioneering self-supervised model, DMASA, which stands out with its dual-masked approach coupled with similarity aggregation for KPI anomaly detection. A key innovation is the integration of spectral residual methods to detect anomalies through frequency-domain analysis, enhancing the identification of their onset points. Furthermore, DMASA refines Transformer architecture to better assimilate the complexities of multivariate time series, while a similarity aggregation technique minimizes interference from irrelevant variables to extract meaningful features from KPIs. The model has proven to be superior and effective across various datasets and can be used in various scenarios.

Our approach is well-suited for detecting anomalies in multivariate KPIs, such as CPU utilization and socket status, in dynamic environments like networks and servers, where conditions are continuously evolving. Traditional methods relying on contrastive learning might struggle with the varying features of data over time. In contrast, our solution, grounded

in mask modeling, is designed to adapt to the data's intrinsic distribution, offering greater flexibility. A key implementation challenge in these contexts is calibrating the anomaly score threshold to ensure timely alerts in response to the evolving landscape. While setting a generalized threshold in advance is difficult due to the varying nature of real-world systems, we believe that dynamically adjusting the threshold based on the statistical distribution of normal data points versus anomalies could significantly improve the model's practicality. Therefore, in future work, we plan to explore how to implement a self-adaptive mechanism for adjusting the threshold at test time.

Additionally, while our current approach effectively identifies when anomalies occur, determining which specific variables contribute to an overall anomaly remains a challenge due to the complex relationships between variables. This topic has not been thoroughly addressed in the field, and existing datasets often lack labels indicating which individual variables are abnormal. In future work, we aim to investigate how the interactions between variables affect anomalies and whether it is possible to attribute an anomaly to specific variables. This will form a key part of our continued research. Meanwhile, we are committed to collecting and incorporating more real-world data to further validate the applicability and robustness of our approach, thereby reinforcing the practical relevance and continuous improvement of our research.

ACRONYMS

FFT	Fast Fourier Transform
KPI	Key Performance Indicator
LSTM	Long Short-term Memory
MAE	Mean Absolute Error
MSL	Mars Science Laboratory rover dataset
PSM	Pooled Server Metrics dataset
RNN	Recurrent Neural Network
SMAP	Soil Moisture Active Passive satellite dataset
SMD	Server Machine Dataset.
SR	Spectral Residual
SWaT	Secure Water Treatment dataset
VAE	Variational Autoencoder

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